

DHRUV MAYUR VIRA

Full-Stack Software Engineer | Backend & Applied AI/ML | Algorithmic Trading Systems

Madhya Pradesh, India | +91 9303000832 | dhruvvira17@gmail.com | github.com/dhruvv16-hash | linkedin.com/in/dhruv-vira-33bb19439

SUMMARY

Computer Science undergraduate (VIT Chennai, Class of 2028) building production full-stack and backend systems: React/Next.js/TypeScript on the frontend, Spring Boot/FastAPI/Node.js REST APIs on the backend, PostgreSQL/Supabase for data. Shipped a multi-tenant SaaS platform with offline-first sync and HMAC-secured payment webhooks, merged production pull requests across 6 open-source organizations (41 public repositories, GitHub Pull Shark badge), and built quantitative-finance and algorithmic-trading systems in Python.

SKILLS

Languages: Java, Python, C++, C, TypeScript, JavaScript, Kotlin

Frontend & Mobile: React.js, Next.js, React Native, Flutter, Vue.js, HTML, CSS, Jest, Cypress

Backend & APIs: Spring Boot, FastAPI, Node.js, REST APIs, Microservices, JDBC, JPA, Webhooks

Databases, Cloud & DevOps: SQL, PostgreSQL, MySQL, Supabase, Firebase, Docker, Kubernetes, Git, GitHub, CI/CD

AI/ML & Quant Finance: Machine Learning, XGBoost, LightGBM, CatBoost, NumPy, Pandas, Scikit-learn, Financial Modeling, Algorithmic Trading

CS Fundamentals: Data Structures & Algorithms (120+ problems solved on LeetCode), OOP, Operating Systems, Computer Networks, Computer Organization & Architecture

PROJECTS

VendorOS — Multi-Tenant Food-Tech SaaS Platform | *Next.js, React, TypeScript, Supabase, PostgreSQL, WhatsApp Cloud API*

- Built a production POS, Kitchen Display System, and inventory-management PWA to replace a restaurant's paper-and-spreadsheet order flow; implemented offline-first sync (localStorage queue → Supabase on reconnect) so staff keep taking and fulfilling orders through connectivity drops with zero order loss.
- Enforced Row-Level Security policies on every Supabase table to isolate tenant data at the database layer, so a single deployment safely serves multiple vendor accounts without per-tenant infrastructure.
- Automated order intake through the Meta WhatsApp Cloud API and secured Razorpay payment webhooks with SHA-256 HMAC signature validation to block forged callbacks; added AI-driven revenue forecasting and stockout alerts to cut manual reorder decisions.

InvestorGPT — Multi-Agent AI Investment Research Platform | *Next.js, FastAPI, Python, SQLAlchemy*

- Architected a 7-agent orchestration pipeline (financial-ingestion, valuation, technical, sentiment, Consensus, and Reviewer agents) that reconciles conflicting buy/sell signals into one research verdict, replacing single-model outputs with a debate-and-consensus workflow.
- Implemented quantitative scoring models — Piotroski F-Score, Altman Z-Score, and DCF sensitivity grids — plus an MPT portfolio studio that simulates 500 Efficient Frontier configurations with Sharpe-ratio optimization to support allocation decisions.

AccidentZero AI — Industrial Safety Risk Monitoring | *Python, FastAPI, XGBoost, LightGBM, CatBoost, LSTM*

- Built a 5-model ensemble (XGBoost, LightGBM, CatBoost, LSTM, rule engine) that scores real-time accident risk from operational metrics, fused with a computer-vision pipeline for PPE-violation and scene-risk detection.
- Exposed predictions via FastAPI batch (Excel) and real-time endpoints with Chart.js dashboards, giving safety officers one interface for retrospective audits and live floor monitoring.

DeltaBridge — Algorithmic Trading Bot (ETH/USD) | *Pine Script, Python, Flask, SQLite, Delta Exchange API, Crypto.com API, Discord/Telegram API*

- Designed a multi-timeframe strategy (UT Bot Alerts + Linear Regression + ADX trend-strength filtering) to suppress false entries from single-indicator systems in choppy price action.
- Built a production Flask bridge from TradingView alerts to Delta Exchange for 24/7 automated execution: multi-threaded daemon polling live candles every 10s, multi-account state, dynamic leverage-adjusted position sizing, trailing SL/TP.
- Secured every order with HMAC-SHA256 signed requests; shipped a live dashboard tracking Win Rate, Profit Factor, and Net PnL with real-time Discord/Telegram trade alerts.

OPEN SOURCE CONTRIBUTIONS & ACHIEVEMENTS

- Winner, iQOO Connect Community Hackathon** — Certificate of Achievement for building and presenting a working prototype under a fixed deadline.
- omegaUp** (Vue.js/TypeScript/PHP) — merged a production PR fixing a stale-state UI bug in the reactive Vue 3 layer; authoring a Google Summer of Code (GSoC) 2026 proposal.
- Supabase** — diagnosed PostgreSQL privilege-leak and DNS-outage issues; built automated diagnostic suites and SQL reconciliation scripts to speed up platform-team triage.
- Kubernetes** (sig-windows) — fixed recurring E2E CI pipeline failures via a Python janitor script that auto-purges soft-deleted Azure AD resources, removing a source of false-negative test runs.
- OpenBB** — merged a PR resolving Poetry packaging conflicts between core modules, eliminating install failures on OS package managers with strict file ownership.
- Kornia** (Computer Vision) — fixed edge-case tensor-validation bugs in the PinholeCamera module, closing a broadcasting failure on unbatched inputs.
- CaPyCLI** (Software Compliance) — built an automated OAuth2 token-refresh lifecycle (Keycloak); refactored auth boilerplate across 16 modules into one unified session manager.

EDUCATION & CERTIFICATIONS

Vellore Institute of Technology, Chennai — B.Tech, Computer Science and Engineering (2024 – 2028)

Coursework: Data Structures & Algorithms, OOP, Computer Networks, Operating Systems, Computer Organization & Architecture, Theory of Computation

Certifications: Microsoft Learn Student Ambassador | Quantitative Research & Software Engineering Virtual Experience — J.P. Morgan Chase & Co. | AI and Data Scientist — One Roadmap